THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Weak forces need little energy.
2	**(a)** 44 g	1 mol CH_4 gives 1 mol CO_2.
3	**(c)** ethene	C_2H_4 = 28.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Incomplete combustion gives soot.
5	**(b)** 36 g	2 mol water = 36 g.
6	**(a)** ethanoic acid	Vinegar.
7	**(c)** A is true but R is false.	A true; in graphite each carbon bonds to three others, so R is false.
8	**(d)** 44.8 L	2 mol CO_2.
9	**(c)** ethanol	Dehydration and oxidation.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Scum forms.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$; 60 g gives 176 g (1); 30 g gives 88 g CO_2 (1).	1+1
12	$3n + 1 = 22$, n = 7 (1); C_7H_{16} (1).	1+1
13	Propyne has a triple bond (1); 2 mol (1).	1+1
14	Molar mass 30 g (1); 24/30 x 100 = 80% (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ratio C : H = 1 : 2, so empirical CH_2 (1); n = 42/14 = 3 (1); C_3H_6, propene (1).	1+1+1
16	n-butane: 3 C-C, 10 C-H (1); isobutane: 3 C-C, 10 C-H (1); Same number of bonds; isomers (1).	1+1+1
17	C1-C4 gases (1); C5-C17 liquids (1); Higher are solids (wax) (1).	1+1+1
	OR: $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ (1); O_2 = 4 mol = 89.6 L (1); CO_2 = 2 mol = 44.8 L (1).	1+1+1
18	Graphite: each C bonded to 3, free electron (1); Diamond: each C bonded to 4, no free electrons (1); Fullerene: discrete C_{60} molecules (1).	1+1+1
	OR: Pentane C_5H_{12}: 72 u (1); Difference CH_2 = 14 u (1); Methane 16, ethane 30, propane 44, butane 58 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dehydration: conc. H_2SO_4, 443 K (1); Oxidation: alkaline $KMnO_4$ (1); $KMnO_4$ is oxidising agent (1); Esterification: conc. H_2SO_4 catalyst (1); Equations of each step (1).	1+1+1+1+1
	OR: Butane and isobutane (1); But-1-ene and but-2-ene (1); Isomers: same formula, different structure (1); Homologues: differ by CH_2 (1); Examples (1).	1+1+1+1+1
20	0.1 mol ethanol (1); $C_2H_5OH + 3\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O$ (1); CO_2 = 0.2 mol = 8.8 g (1); Water = 0.3 mol = 5.4 g (1); Volume = 4.48 L (1).	1+1+1+1+1
	OR: Diamond: 3D, hard (1); Graphite: layered, conducts (1); Fullerene: C_{60} (1); All burn to CO_2 (1); Same element (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 88 g	1 mol gives 2 mol CO_2.
2	**(c)** methyl ethanoate	Ester of ethanoic acid.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Partial ionisation.
4	**(c)** 75%	12/16 x 100 = 75%.
5	**(a)** unsaturated	High carbon percentage.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Both explain the variety.
7	**(b)** 85.7%	24/28 = 85.7%.
8	**(d)** hydrocarbon tails	Tails cluster with grease.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The pi bond opens.
10	**(d)** 92.3%	24/26 = 92.3%.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ethyl ethanoate (1); Fruity smell (1).	1+1
12	Methane has 75% C, ethyne 92% (1); Limited oxygen leaves unburnt carbon (1).	1+1
13	Carbon is not fully oxidised (1); Ethyne (highest carbon %) (1).	1+1
14	Lower energy per litre than petrol (1); Advantage: burns cleanly, renewable (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$, addition (1); $CH_3CH_3 + Cl_2 \longrightarrow CH_3CH_2Cl + HCl$ (1); Substitution (sunlight) (1).	1+1+1
16	Ethyl ethanoate (1); Alcohol: ethanol (1); $CH_3COOC_2H_5 + H_2O \longrightarrow CH_3COOH + C_2H_5OH$ (1).	1+1+1
17	Hydration of ethene (1); $CH_2{=}CH_2 + H_2O \longrightarrow CH_3CH_2OH$ (1); Acid catalyst (1).	1+1+1
	OR: X: ethanoic acid (1); Product: ethyl ethanoate (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1).	1+1+1
18	Bromine water: ethene decolourises (1); Sodium bicarbonate: acid effervesces (1); Ethane and ethanol do not (1).	1+1+1
	OR: Do not form scum (1); Their salts with Ca/Mg are soluble (1); Work in acidic water too (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A: ethanol (1); B: ethanoic acid (1); C: ethyl ethanoate (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Oxidation and esterification (1).	1+1+1+1+1
	OR: Methane 75% (1); Ethene 85.7% (1); Ethyne 92.3% (1); Higher carbon gives more soot (1); Ethyne most sooty (1).	1+1+1+1+1
20	Soap: $C_{17}H_{35}COONa$ (1); Detergent: sulphonate salt (1); Micelle: tails inward (1); Hard water gives scum with soap (1); Detergent works in hard water (1).	1+1+1+1+1
	OR: $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$ (1); Na2CO3 = 106 g/mol, so 0.1 mol (1); Acid = 0.2 mol = 12 g (1); CO_2 = 0.1 mol (1); Volume 2.24 L (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** ethane	No acidic hydrogen.
2	**(c)** A is true but R is false.	A true; R false (it is saturated).
3	**(c)** 2	Triple bond to single needs 2 H_2.
4	**(c)** ethyne	Triple bond is shortest.
5	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but R does not explain the smell.
6	**(d)** 3	n-, iso-, neopentane.
7	**(a)** ethanol is oxidised and $KMnO_4$ is reduced	$KMnO_4$ acts as oxidising agent.
8	**(a)** Both A and R are true and R is the correct explanation of A.	Functional group governs reactions.
9	**(b)** 5	Hexane isomers.
10	**(c)** ester (-COO-)	Ester.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hydrogen (1); Burns with a pop sound (1).	1+1
12	Ethanol forms hydrogen bonds with water (1); Hexane is non-polar (1).	1+1
13	Soap is made from fats or vegetable oil with alkali (1); Mineral oil is hydrocarbon with no ester to hydrolyse (1).	1+1
14	Ethanol (1); Hydrogen bonding between -OH groups (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Saponification needs a base (1); Hydroxide attacks the ester (1); Gives soap (salt) which cannot reverse (1).	1+1+1
16	Removing four electrons needs huge energy (1); Gaining four electrons is also unfavourable (1); Carbon shares electrons (1).	1+1+1
17	Fat + NaOH gives soap + glycerol (1); Soap is a sodium salt of fatty acid (1); Salt reduces solubility so soap separates (1).	1+1+1
	OR: Distilled water: lather (1); Hard water: scum, little lather (1); Ca/Mg ions form insoluble salts (1).	1+1+1
18	Functional group decides reactions (1); -OH: alcohol reactions (1); -COOH: acidic character (1).	1+1+1
	OR: Acid forms dimers via hydrogen bonds (1); Stronger attractions (1); Ethanoic acid 391 K vs ethanol 351 K (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	General formulas $C_nH_{2n}n_2^+$, C_nH_{2n}, C_nH_{2n}-2 (1); Single, double, triple bonds (1); Alkanes: substitution; others: addition (1); Alkenes and alkynes decolourise bromine water (1); Examples: ethane, ethene, ethyne (1).	1+1+1+1+1
	OR: Neutralisation gives salt and water (1); Carbonate gives CO_2 (1); Alcohol gives ester (1); Uses: preservative (1); Esters: perfumes (1).	1+1+1+1+1
20	X: ethene (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (1); Product: ethane (1); Ni is catalyst (1); Ethane has lower C% so burns cleanly (1).	1+1+1+1+1
	OR: Litmus: ethanoic acid red (1); Bicarbonate: ethanoic acid effervesces (1); Bromine water: ethene decolourised (1); Sodium: ethanol releases hydrogen (1); Ethane is left (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.